

## SUPPORT FOR THE SURVEY SPONSOR AND NONRESPONSE BIAS

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**Abstract** In an experiment designed to examine nonresponse bias, either the March of Dimes or the University of Michigan was identified as the sponsor of a survey mailed to individuals whose level of support for the March of Dimes was known. The response rate was higher to the university survey, but support for the March of Dimes increased survey participation to the same extent in both conditions. As a result of the overrepresentation of supporters of the organization, both surveys showed nonresponse bias for variables linked to support. The bias was greater, however, when the sponsor was identified as the March of Dimes. Thus, the university sponsor brought in not only more of the

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sample but also a more representative sample on variables related to support for the March of Dimes. Overall, the magnitude of the relationship between support for the organization and nonresponse was not a strong predictor of the magnitude of the nonresponse bias. The results demonstrate that the simple “common cause” model of nonresponse will not always apply, and that the model should be extended to incorporate multiple auxiliary variables.

In the past decade, research has demonstrated that nonresponse rates and nonresponse bias are often unrelated (Keeter et al. 2000; Groves and Peytcheva 2008). Thus, focusing on minimizing nonresponse rates as *the* way to reduce nonresponse bias is ill conceived. Devising other methods to reduce nonresponse bias depends on understanding the mechanisms that can link nonresponse rates to nonresponse bias.

Nonresponse bias occurs when there is a connection between the likelihood of participation in the survey and the specific survey measure of interest. This observation is conveyed in Bethlehem's (2002) expression of the nonresponse bias of an unadjusted respondent mean as  $\frac{\sigma_{YP}}{\bar{P}}$ , where  $\sigma_{YP}$  is the covariance between a survey variable  $Y$  and the probability of participating in the survey,  $P$ , and  $\bar{P}$  is the mean probability of participating in the survey, each measured on the entire population (see also Lessler and Kalsbeek 1992, Chapter 7). In terms of correlations, this may be expressed as  $\frac{\rho_{YP}}{\bar{P}}(\sigma_Y\sigma_P)$ , where  $\sigma_Y$  and  $\sigma_P$  are the population standard deviations of the  $Y$  and  $P$  variables, respectively. These expressions remind us that nonresponse bias can be large for some  $Y$  variables and small for others. Thus, research should attend to identifying the circumstances that produce high covariances (or correlations) versus those that produce low ones.

One way the covariance term can be large is if survey participation and the survey variable share the same cause—a circumstance Groves (2006) describes as the common-cause model. When there is such a common causal variable, the data for the survey variable will be, in Little and Rubin's (2002) terms, “missing at random,” that is, conditional on the value of the shared causal variable ( $Z$ ), there will be no nonresponse bias for the  $Y$  variable. If the common cause fully accounts for the correlation of  $P$  and  $Y$ , then that correlation can be expressed as the product of the two correlations involving  $Z$ :  $\rho_{ZP} \times \rho_{ZY}$ . This is good news for those worried about nonresponse bias. The correlation of  $Z$  with  $P$  and the correlation of  $Z$  with  $Y$  both have to be high for the bias to be large.

Interest in the survey topic is probably the most often studied potential common cause of survey participation and survey responses. Baumgartner and Rathbun (1997) and Groves, Presser, and Dipko (2004) showed how nonresponse bias arises when interest in the topic is a motivation for participation *and* a given estimate is based on questions that measure (or correlate with) that interest.

Another potential common cause is attitude toward the survey sponsor. Such attitudes undoubtedly play a role in the tendency for government survey organizations to obtain higher cooperation rates than scientific organizations, which in turn tend to obtain higher cooperation rates than commercial groups. Of course, government, academic, and commercial surveys differ in many ways in addition to sponsorship, but the difference in response rates has been found in experimental comparisons that hold constant other features (Presser, Blair, and Triplett 1992).

The effects of feelings toward the sponsoring organization may be enhanced when the organization has provided benefits to individuals, possibly triggering a desire to reciprocate. Moreover, people may derive intangible benefits from such acts of reciprocation, for example a sense of social solidarity (with regard to a voluntary organization) or fulfillment of civic duty (in connection with a government organization). As suggested by leverage-salience theory (Groves, Singer, and Corning 2000), these attitudinal states can increase the likelihood of participation when the sponsor of the survey is positively valued *and* the sponsorship is made salient during the survey request. Thus, support for a sponsoring organization of a survey may increase participation rates. In these cases, nonresponse bias seems likely to result for items that are correlates of the support for the organization.

This line of thinking underlies the widespread practice of using third-party sponsors for customer satisfaction surveys (instead of the service provider itself). Likewise, the presumed link between attitudes toward the sponsor and response rates underlies the expectation that surveys sponsored by a political party will generate higher estimates of support for its candidates and that surveys of an organization's members conducted by the organization will have more participation from committed members than from casual members. However, we know of no experimental research that demonstrates the impact of support for the sponsor on survey response rates.

In this article, we address two research questions: 1) Does survey sponsorship influence response propensities in line with the sponsorship effect hypothesis? and 2) Under what circumstances is support for the sponsor an important predictor of nonresponse bias?

## Research Design

We mounted a between-subjects randomized experiment that varied sponsorship and topic. Three different questionnaires were mailed in 2007 to random subsamples of individuals listed in a nationwide database from that year maintained by the March of Dimes. One questionnaire used the March of Dimes logo and was titled "The March of Dimes Survey of Giving and Volunteering"; a second questionnaire included the same questions but used the University of Michigan logo and was titled "The University of Michigan Survey of Giving

and Volunteering”; a final questionnaire included many of the same questions as the other two versions but used the University of Michigan logo and was titled “The University of Michigan Survey of Jobs and the Labor Force.” The mailing protocol for each survey consisted of an advance letter, a paper questionnaire package (containing a cover letter, a twelve-page questionnaire, and a postpaid return envelope), a reminder postcard, and one follow-up questionnaire package for nonrespondents.

The March of Dimes is a nonprofit organization devoted to preventing birth defects, premature birth, and infant mortality. It carries out this mission through research, community services, education, and advocacy. Prior to sampling, we discovered that the majority of the entries in the March of Dimes database were individuals who had merely been solicited for contributions. Only a small minority had given donations. As a result, we selected the sample using stratification to give more equal representation to individuals with differing levels of engagement with the organization.

The strata were created using several indicators of support—number, size and recency of donations, and extent of volunteering activity. We combined these indicators to create an index with six levels, ranging from those who never contributed through those who made multiple donations and participated in activities, as shown in table 1.

Exactly 1,000 individuals were selected at random from each level of the support index, yielding a total sample of 6,000. Because our intent was not to estimate parameters of the overall population represented by the March of Dimes frame but instead to test the relationship between organizational support and both survey cooperation and nonresponse bias, we did not use weights to adjust for the unequal probabilities of selection. We did, however, examine the effect of weights constructed to adjust for nonresponse differences across the strata.

## **Does the Sponsorship of the Survey Affect Response Propensities?**

As shown in the bottom row of table 2, the survey topic had no effect on cooperation, but sponsorship did. When the sponsor was identified as the University of Michigan, the volunteering survey had a response rate almost identical to that of the labor force survey (23 percent versus 24 percent). By contrast, despite the fact that the samples contained many March of Dimes supporters, the response rate to the University of Michigan volunteering survey was nearly twice the response rate to the March of Dimes volunteering survey (23 percent versus 12 percent). Although counter to our initial hypothesis, our post hoc interpretation is that mail from the March of Dimes was more apt to make people think “fund-raising” (and thus was less likely to be opened) than mail from the University of Michigan.

**Table 1. Dual-Purpose Sampling Strata and Support Index**

| Stratum | Description                                                                                                                                                                                           | Recency of donation | Size of donation              | Multiple donor                    |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-------------------------------|-----------------------------------|
| 1       | Individuals who never donated                                                                                                                                                                         | None                | None                          | No                                |
| 2       | All donors who do not fall into any of Strata 1, 3, 4, 5, or 6                                                                                                                                        | Varies              | Varies                        | No                                |
| 3       | All single donors (no multi-donors) who are in either of the following groups: (0–12 months' recency of donation and less than \$50 donated) or (13–24 months' recency of donation and \$50+ donated) | Varies              | Varies                        | Only single donors                |
| 4       | All multiple donors excluding anyone in strata 5 or 6                                                                                                                                                 | Varies              | Varies                        | Only multiple donors              |
| 5       | 0–12 months' donation recency and greater than or equal to \$50 donation (includes multi-buyer and single buyer); however, this group excludes anyone in stratum 6                                    | Less than 12 months | Greater than or equal to \$50 | Both multi-donor and single donor |
| 6       | Mothers March volunteers as well as member of one or more of the following: DM, IN, MM donor, walk donor, or walker                                                                                   | Varies              | Varies                        | Multi-donors                      |

Did degree of commitment to the organization influence the response rate in line with the sponsorship hypothesis? Table 2 shows that response rates increased as a function of the support index. However, this was true not only for the March of Dimes condition, but for both University of Michigan conditions as well. In a logistic regression model predicting the probability of returning a questionnaire, the interaction between a three-category factor defining combinations of sponsorship and topic (March of Dimes—Volunteering, University of Michigan—Volunteering, or University of Michigan—Labor Force) and the support index was not significant, but each of the main effects (sponsorship/topic and support) was highly significant. We interpret the support effect as reflecting a tendency for those who give to charitable causes to be more likely to consent to survey requests in general, even those unrelated

**Table 2. Percentage of Mail Questionnaires Completed and Returned by Topic/Sponsor and Support for the March of Dimes** (numbers in parentheses represent cell sample sizes)

| Support index<br>(strata) | Volunteering survey            |                                           | Labor force<br>survey                     |               | TOTAL |
|---------------------------|--------------------------------|-------------------------------------------|-------------------------------------------|---------------|-------|
|                           | Sponsored by<br>March of Dimes | Sponsored by<br>University of<br>Michigan | Sponsored by<br>University of<br>Michigan |               |       |
| 1—lowest                  | 7.7% (324)                     | 17.1% (322)                               | 21.2% (354)                               | 15.5% (1,000) |       |
| 2                         | 8.3% (348)                     | 18.1% (337)                               | 18.7% (315)                               | 14.9% (1,000) |       |
| 3                         | 14.0% (365)                    | 25.3% (324)                               | 23.8% (311)                               | 20.7% (1,000) |       |
| 4                         | 10.6% (330)                    | 23.6% (331)                               | 25.4% (339)                               | 19.9% (1,000) |       |
| 5                         | 14.7% (319)                    | 26.7% (344)                               | 23.7% (337)                               | 21.9% (1,000) |       |
| 6—highest                 | 19.2% (339)                    | 28.7% (331)                               | 30.3% (330)                               | 26.0% (1,000) |       |
| TOTAL                     | 12.4% (2,025)                  | 23.3% (1,989)                             | 23.9% (1,986)                             | 19.8% (6,000) |       |

NOTE.—All *N*s represent the number of questionnaires mailed out.

to charitable activities. This is consistent with prior findings that individuals who had done volunteer work were more apt to become survey respondents (Abraham, Helms, and Presser 2009; Groves, Singer, and Corning 2000) and that voters and those who made campaign contributions were more likely than others to become survey respondents, even when the survey was not about politics (Tourangeau, Groves, and Redline 2010; Groves, Presser, and Dipko 2004).

**Under What Circumstances Is Support for the Sponsor a Predictor of Nonresponse Bias in Respondent Estimates?**

Although sponsorship did not moderate the relationship between support and response rate, estimates for respondents on items related to the support index are likely to show nonresponse bias, since respondents (for both survey topics and sponsors) were disproportionately individuals with higher levels of support for the March of Dimes. Table 3 confirms this expectation for nine frame variables related to the support index, for each combination of sponsor and topic. In nearly every case, the respondents show higher levels of support than the initial total sample.

However, the magnitude of this nonresponse bias is larger for the March of Dimes condition than for the University of Michigan conditions. In other words, although those committed to the March of Dimes are overrepresented in all the surveys, the overrepresentation is particularly acute with the March of Dimes sponsorship. Thus, the university sponsor brings in not only more

**Table 3. Respondent Means, Nonresponse Adjusted Means, and Full Sample Means for Nine Frame Variables Related to Support, by Sponsor/Topic (standard errors in parentheses)**

| (1)<br>Sampling frame variable                                       | (2)<br>Sponsor/topic | (3)<br>Respondent<br>mean (SRS SE) | (4)<br>Adjusted respondent<br>mean (linearized SE) | (5)<br>Full sample<br>mean (SRS SE) |
|----------------------------------------------------------------------|----------------------|------------------------------------|----------------------------------------------------|-------------------------------------|
| Last volunteer kit dollar amount (log-transformed)                   | MODV                 | 0.91 (0.10)                        | 0.59 (0.07)                                        | 0.57 (0.03)                         |
|                                                                      | UMV                  | 0.72 (0.07)                        | 0.58 (0.06)                                        | 0.57 (0.03)                         |
|                                                                      | UMLF                 | 0.73 (0.07)                        | 0.58 (0.06)                                        | 0.57 (0.03)                         |
| Number of times unfulfilled                                          | MODV                 | 0.12 (0.02)                        | 0.19 (0.04)                                        | 0.20 (0.01)                         |
|                                                                      | UMV                  | 0.15 (0.02)                        | 0.20 (0.03)                                        | 0.20 (0.01)                         |
|                                                                      | UMLF                 | 0.17 (0.02)                        | 0.19 (0.02)                                        | 0.22 (0.01)                         |
| Lifetime total times volunteered                                     | MODV                 | 0.61 (0.09)                        | 0.40 (0.06)                                        | 0.29 (0.02)                         |
|                                                                      | UMV                  | 0.39 (0.05)                        | 0.32 (0.04)                                        | 0.28 (0.02)                         |
|                                                                      | UMLF                 | 0.36 (0.04)                        | 0.29 (0.03)                                        | 0.28 (0.02)                         |
| Lifetime total amount raised volunteering more than \$50 (indicator) | MODV                 | 0.14 (0.02)                        | 0.09 (0.02)                                        | 0.07 (0.01)                         |
|                                                                      | UMV                  | 0.11 (0.01)                        | 0.09 (0.01)                                        | 0.07 (0.01)                         |
|                                                                      | UMLF                 | 0.08 (0.01)                        | 0.07 (0.01)                                        | 0.06 (0.01)                         |
| Lifetime donation amount more than \$50 (indicator)                  | MODV                 | 0.37 (0.03)                        | 0.34 (0.03)                                        | 0.31 (0.01)                         |
|                                                                      | UMV                  | 0.37 (0.02)                        | 0.35 (0.02)                                        | 0.31 (0.01)                         |
|                                                                      | UMLF                 | 0.33 (0.02)                        | 0.32 (0.02)                                        | 0.32 (0.01)                         |

*Continued*

Table 3. Continued

| (1)<br>Sampling frame variable                                   | (2)<br>Sponsor/topic | (3)<br>Respondent<br>mean (SRS SE) | (4)<br>Adjusted respondent<br>mean (linearized SE) | (5)<br>Full sample<br>mean (SRS SE) |
|------------------------------------------------------------------|----------------------|------------------------------------|----------------------------------------------------|-------------------------------------|
| Number of lifetime donations (log-transformed)                   | MODV                 | 1.40 (0.05)                        | 1.38 (0.05)                                        | 1.29 (0.02)                         |
|                                                                  | UMV                  | 1.38 (0.04)                        | 1.35 (0.04)                                        | 1.30 (0.02)                         |
|                                                                  | UMLF                 | 1.32 (0.03)                        | 1.29 (0.03)                                        | 1.28 (0.02)                         |
| Last year money raised more than \$50 (indicator)                | MODV                 | 0.06 (0.02)                        | 0.05 (0.01)                                        | 0.02 (<0.01)                        |
|                                                                  | UMV                  | 0.02 (0.01)                        | 0.01 (<0.01)                                       | 0.02 (<0.01)                        |
|                                                                  | UMLF                 | 0.02 (0.01)                        | 0.02 (0.01)                                        | 0.02 (<0.01)                        |
| Lifetime total amount raised at walks more than \$50 (indicator) | MODV                 | 0.06 (0.01)                        | 0.05 (0.01)                                        | 0.02 (<0.01)                        |
|                                                                  | UMV                  | 0.02 (0.01)                        | 0.02 (0.01)                                        | 0.02 (<0.01)                        |
|                                                                  | UMLF                 | 0.03 (0.01)                        | 0.04 (0.01)                                        | 0.03 (<0.01)                        |
| Total dollars raised lifetime walks (log-transformed)            | MODV                 | 0.37 (0.09)                        | 0.37 (0.09)                                        | 0.17 (0.02)                         |
|                                                                  | UMV                  | 0.18 (0.05)                        | 0.18 (0.05)                                        | 0.15 (0.02)                         |
|                                                                  | UMLF                 | 0.21 (0.05)                        | 0.21 (0.05)                                        | 0.18 (0.02)                         |

NOTE.—Cells contain estimated means (and standard errors of estimates) for MOD—Volunteering (MODV;  $n = 252$ ), UM—Volunteering (UMV;  $n = 463$ ), and UM—Labor Force (UMLF;  $n = 474$ ) conditions. Natural-log transformations were applied to three variables to reduce the effects of extreme values on the estimates. For these variables, a value of 1 was added to the respondent value so that the transformation could be performed for cases with a respondent value of 0.

of the sample but also a more representative set of the initial sample on the support variables.

Table 3 also shows the extent to which the bias can be offset by nonresponse adjustment weights that take into account the differences in response rates across the six levels of support. Comparing the adjusted estimates for respondents (column 4) with the full sample estimates (column 5) indicates that the adjustments repair most (but not all) of the bias in the unweighted respondent estimates (column 3). These results, as well as the tendency for the weighted standard errors to be smaller than the unweighted standard errors, are consistent with theoretical expectations (Little and Vartivarian 2005). But, of course, most surveys do not have access to our kind of frame variables for weighting adjustments.

Is the magnitude of the nonresponse bias for an item predicted by the correlation between the item and propensity to respond to the survey? The rows of table 4 contain the variables appearing on the March of Dimes sampling frame, ordered by the absolute (i.e., unsigned) value of the variable's correlation with the support index, ranging from a high of 0.627 for the last volunteer-kit dollar amount (log-transformed) to a low of 0.055 for lifetime amount raised at walks (log-transformed). If support were the single common cause of the survey variable and response propensity, the correlation  $\rho_{YP}$ , which appears in the numerator of the nonresponse bias expression  $\frac{\rho_{YP}}{P}(\sigma_Y\sigma_P)$ , could be estimated by the product of  $\rho_{ZP} \times \rho_{ZY}$  (the product of the two paths in the single common-cause model), where  $Z$  is the support index,  $Y$  is the given frame variable, and  $P$  is the response propensity (represented by the indicator of responding to the survey request). In our sample,  $\rho_{ZP}$  is 0.10. We multiplied 0.10 by the correlation in the second column of table 4 to produce the third column, thereby providing an indication of how the connection of support to both response propensity and the frame variable acts to dampen the nonresponse bias. In other words, the expected correlation of propensity and the frame variable is only 10 percent of the correlation between the support index and the frame variable.

How useful is the expected correlation between the frame variable and response propensity under the single common-cause model in predicting the levels of nonresponse bias in estimates? Only moderately so. For example, the last volunteer-kit dollar amount (log-transformed), the frame variable with the highest absolute correlation to the support index (0.627), has a relative bias of about 38 percent of the full sample estimate, with respondents (as shown in table 3) having somewhat higher dollar amounts than nonrespondents. By contrast, raising more than \$50 in the past year, which has a relatively low correlation with the support index (0.117), has the largest relative bias—113 percent of the sample estimate—indicating higher proportions raising \$50 in the past year among respondents as compared with nonrespondents (especially, as shown in table 3, among respondents to the March of Dimes questionnaire). Overall, the correlation between the expected correlation of the frame variable

Table 4. Correlations of the Frame Variables with the Support Index and Response Propensity (Expected and Actual), along with the Ratios of the Nonresponse Bias to the Full Sample Estimate (for the Two Volunteering Surveys)

| (1)<br>Sampling frame variables                               | Correlation of frame variable with ... |                                        |                                      | (5)<br>Ratio of nonresponse bias to full sample estimate*** |
|---------------------------------------------------------------|----------------------------------------|----------------------------------------|--------------------------------------|-------------------------------------------------------------|
|                                                               | (2)<br>Support index                   | (3)<br>Response propensity (expected)* | (4)<br>Response propensity(actual)** |                                                             |
| Last volunteer kit amount (log-transformed, mean)             | 0.627                                  | 0.063                                  | 0.077                                | 0.379                                                       |
| Number of times unfulfilled (mean)                            | -0.560                                 | -0.056                                 | -0.055                               | -0.313                                                      |
| Lifetime total times volunteered (mean)                       | 0.502                                  | 0.050                                  | 0.105                                | 0.649                                                       |
| Lifetime total amount raised volunteering more than \$50      | 0.387                                  | 0.039                                  | 0.099                                | 0.794                                                       |
| Lifetime donation amount more than \$50                       | 0.226                                  | 0.023                                  | 0.066                                | 0.212                                                       |
| Lifetime donations (log-transformed, mean)                    | 0.123                                  | 0.012                                  | 0.062                                | 0.075                                                       |
| Last year money raised more than \$50                         | 0.117                                  | 0.012                                  | 0.063                                | 1.133                                                       |
| Lifetime total amount raised at walks more than \$50          | 0.059                                  | 0.006                                  | 0.040                                | 0.619                                                       |
| Lifetime total amount raised at walks (log-transformed, mean) | 0.055                                  | 0.006                                  | 0.048                                | 0.582                                                       |

\*Expected response propensity is the correlation of the frame variable with the support index (column 2) multiplied by the correlation of the support index with whether or not the sample member completed a questionnaire (which is 0.10).

\*\*Actual response propensity is the correlation of the frame variable with whether or not the sample member completed a questionnaire.

\*\*\*This is the respondent mean (across both conditions) minus the full sample mean (across both conditions), divided by the full sample mean (across both conditions). Therefore, in the case of the number of times unfulfilled, the respondent mean is biased low, and the bias (in absolute value) is only about 31 percent of the full sample estimate. Larger absolute values indicate larger biases relative to the full sample estimate.

with the response propensity, on the one hand, and the relative nonresponse bias, on the other (that is, between columns 3 and 5 in table 4), is 0.52.

Why is the expected correlation of the support index with response propensity under the single common-cause model not a stronger predictor of nonresponse bias? Because other factors related to the frame variables also affect response propensity. This is shown in the fourth column of table 4, which presents the actual correlations between the frame variables and response propensity. These values are a function of all the shared influences on response propensity and the frame variables, measured or unmeasured, not just those captured in our support index.

The actual correlations of the frame variables with response propensity (column 4) are typically larger (by varying magnitudes) and in one case slightly smaller than the expected correlations of the same frame variables with response propensity under the single common-cause model (column 3). This means that additional correlates of response propensity and the frame variables can combine with the support variable and create either more or less susceptibility to bias. Furthermore, as expected from the bias expression given by Bethlehem (2002), the actual correlations in column 4 are stronger predictors of the level of nonresponse bias (column 5): The correlation between these two columns is 0.65. Thus, the actual correlations explain over 50 percent more of the variation in the bias of estimates than do the correlations expected under the single common-cause model—42 percent versus 27 percent.

In summary, support for the sponsor as measured by our index appears to be one variable that produces nonresponse bias in the estimated means of items related to the sponsor. If we ask, however, whether support for the sponsor, by itself, is predictive of nonresponse bias, the answer depends on the relative impact of all the other influences on response propensity and the survey item. Although support for the sponsor can produce a link between response rates and the values of some of the survey variables, other factors can act to suppress this source of nonresponse bias because of their own impacts on response propensity, on the survey item, or on both. (For similar results based on simulated data, see Kreuter and Olson [2011].)

## Discussion

As is true for all research, our results may be conditional on the set of procedures we used. Our procedures were similar to those used in many mail surveys, but some postal surveys use more intensive follow-ups than we did. Although we think it unlikely, it is possible that greater nonresponse follow-ups would not only have decreased the nonresponse rate, but also differently affected the composition of the nonrespondents in the two sponsorship conditions. More generally, replication of our experiment with other topics, sponsors, and survey modes would be desirable.

Although we were unable to verify the hypothesis that support for the March of Dimes would lead to higher rates of participation in a March of Dimes survey than to the same survey sponsored by the University of Michigan, our test was confounded by the fact that one of the sponsors was generally associated with charitable solicitations, whereas the other was not. However, regardless of who was identified as the survey sponsor, increased support for the March of Dimes was related to increased survey participation. As a result, we found that the surveys' estimates of means for variables linked to support for the organization were biased by the overrepresentation of respondents who had supported the March of Dimes in the past. But this bias in support for the March of Dimes was greater when that organization was identified as the survey sponsor.

The magnitude of the relationship between support for the organization and nonresponse rate was not a strong predictor of the magnitude of the nonresponse bias for estimates related to support. Even though we demonstrated the importance of support ( $Z$ ) for both response propensity and some  $Y$  variables, nonresponse bias of an estimated mean of  $Y$  depended on how other variables ( $Z$ 's) combined with support to jointly affect the likelihood of participation ( $P$ ) and  $Y$ . Our results (consistent with the results of Kreuter and Olson's [2011] simulations) show that the simple "common-cause" model (Groves 2006) will not always apply, and thus the model ought to be extended to incorporate multiple auxiliary variables. Future research should aim to identify the patterns of associations among multiple auxiliary variables, response propensity, and key survey variables that result in nonresponse bias.

In practice, the sponsorship of a survey is rarely considered as a design feature subject to manipulation by the researcher. Our findings suggest that more attention might usefully be paid to this decision. When the sponsor has a well-known point of view on a topic or when the sponsor provides benefits or enforces regulations related to the survey topic, nonresponse bias in survey estimates based on items related to the sponsor may result. In such cases, either alternative sponsors without such linkages should be considered or attempts to deemphasize the sponsorship in recruitment should be attempted (with appropriate attention to informed consent). Our findings suggest that if the survey design offers other reasons to respond that compete with survey sponsorship as a mechanism for response propensity, they will reduce or eliminate the biasing impact of support for the sponsor.

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